

Fourier Analysis on the Circle: Uniqueness, Convolution, and Summability

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May 4, 2026

Abstract

In this paper, we study several foundational ideas in Fourier analysis on the circle. We begin by asking a basic question of uniqueness: if all of the Fourier coefficients of an integrable function vanish, to what extent must the function itself vanish? We then introduce convolution as a natural way to rewrite Fourier partial sums and to organize many of the main arguments that follow. From there, we define good kernels and show how their concentration properties allow us to recover a function from suitable averages.

We next apply this framework to two important summability methods. First, we study Cesàro summability through the Fejér kernel, which gives a robust form of convergence even when ordinary convergence of Fourier series is delicate. We then turn to Abel summability and the Poisson kernel, which provide another method for recovering a function from its Fourier coefficients. Finally, we use the Poisson kernel to construct harmonic extensions of boundary data and connect these ideas to the Dirichlet problem in the unit disc.

We emphasize how uniqueness leads naturally to convolution, convolution leads to kernels, and kernels lead to summability and harmonic extension.

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Introduction

Rather than studying a periodic function directly, we study the sequence of its Fourier coefficients and ask how much of the original function can be recovered from that frequency data. This leads to several natural questions. When do the Fourier coefficients determine the function uniquely? How should we interpret the partial sums of the Fourier series? And what can be done when ordinary convergence is too delicate to handle directly?

We develop a sequence of ideas that address those questions. We begin with a uniqueness result showing that if all Fourier coefficients of an integrable function vanish, then the function itself must vanish at each point of continuity. We then introduce convolution, which gives a convenient way to rewrite Fourier partial sums in terms of kernels. This point of view leads naturally to the notion of a good kernel, whose concentration properties make it possible to recover a function from suitable averages.

We then apply this framework to two important summability methods. The Fejér kernel gives Cesàro summability of Fourier series, while the Poisson kernel gives Abel summability. These methods show that even when ordinary convergence of Fourier series is subtle, one can still recover the original function in a robust sense. Finally, the Poisson kernel allows us to pass from Fourier series on the circle to harmonic functions in the unit disc, leading to the Dirichlet problem and its solution by Poisson integrals.

The first theorem begins with the most basic uniqueness question. The proof of the first theorem introduces a method that appears throughout the paper, namely testing a function against trigonometric polynomials or kernels that concentrate near a point.

1 Fourier Analysis on the Circle

Theorem 1.1. *Suppose that f is an integrable function on the circle with*

$$\widehat{f}(n) = 0 \quad \text{for all } n \in \mathbb{Z}.$$

Then $f(\theta_0) = 0$ whenever f is continuous at the point θ_0 .

Proof. We first deal with the case in which f is real-valued. We argue by contradiction. Assume there is a point θ_0 at which f is continuous and $f(\theta_0) \neq 0$. Replacing f by $-f$ if necessary, we may assume that

$$f(\theta_0) > 0.$$

By translating the variable, we may also assume without loss of generality that

$$\theta_0 = 0.$$

Thus f is continuous at 0 and

$$f(0) > 0.$$

Our goal is to construct a sequence of trigonometric polynomials $\{p_k\}$ which become increasingly concentrated near 0, in such a way that

$$\int_{-\pi}^{\pi} f(\theta) p_k(\theta) d\theta \rightarrow \infty \quad \text{as } k \rightarrow \infty.$$

This will contradict the hypothesis that all Fourier coefficients of f vanish.

Since f is continuous at 0 and $f(0) > 0$, we can choose δ with

$$0 < \delta \leq \frac{\pi}{2}$$

such that

$$f(\theta) > \frac{f(0)}{2} \quad \text{whenever } |\theta| < \delta.$$

Now define

$$p(\theta) = \varepsilon + \cos \theta,$$

where $\varepsilon > 0$ is chosen small enough so that

$$|p(\theta)| < 1 - \frac{\varepsilon}{2} \quad \text{whenever } \delta \leq |\theta| \leq \pi.$$

Next choose η with

$$0 < \eta < \delta$$

such that

$$p(\theta) \geq 1 + \frac{\varepsilon}{2} \quad \text{whenever } |\theta| < \eta.$$

For each $k \geq 1$, let

$$p_k(\theta) = [p(\theta)]^k.$$

Since p is a trigonometric polynomial, each p_k is also a trigonometric polynomial.

Since $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, the integral of f against any trigonometric polynomial must vanish. Therefore,

$$\int_{-\pi}^{\pi} f(\theta) p_k(\theta) d\theta = 0 \quad \text{for every } k.$$

On the set $\{\theta : \delta \leq |\theta| \leq \pi\}$, we have

$$|p(\theta)| \leq 1 - \frac{\varepsilon}{2},$$

hence

$$|p_k(\theta)| \leq \left(1 - \frac{\varepsilon}{2}\right)^k.$$

Therefore,

$$\left| \int_{\delta \leq |\theta| \leq \pi} f(\theta) p_k(\theta) d\theta \right| \leq \left(1 - \frac{\varepsilon}{2}\right)^k \int_{-\pi}^{\pi} |f(\theta)| d\theta.$$

On the other hand, for $|\theta| < \eta$ we have both

$$f(\theta) \geq \frac{f(0)}{2} \quad \text{and} \quad p_k(\theta) \geq \left(1 + \frac{\varepsilon}{2}\right)^k.$$

Hence

$$\int_{|\theta| < \eta} f(\theta) p_k(\theta) d\theta \geq 2\eta \cdot \frac{f(0)}{2} \left(1 + \frac{\varepsilon}{2}\right)^k. \tag{1}$$

Also, by our choice of δ , both $f(\theta)$ and $p_k(\theta)$ are nonnegative whenever $|\theta| < \delta$, so

$$\int_{\eta \leq |\theta| < \delta} f(\theta) p_k(\theta) d\theta \geq 0. \quad (2)$$

Therefore,

$$\int_{|\theta| < \delta} f(\theta) p_k(\theta) d\theta \geq 2\eta \cdot \frac{f(0)}{2} \left(1 + \frac{\varepsilon}{2}\right)^k. \quad (3)$$

Combining the estimates (1), (2), and (3), together with the bound on the complementary region, we obtain

$$\int_{-\pi}^{\pi} f(\theta) p_k(\theta) d\theta \geq 2\eta \cdot \frac{f(0)}{2} \left(1 + \frac{\varepsilon}{2}\right)^k - \left(1 - \frac{\varepsilon}{2}\right)^k \int_{-\pi}^{\pi} |f(\theta)| d\theta.$$

As $k \rightarrow \infty$, the right-hand side tends to ∞ , which contradicts the fact that

$$\int_{-\pi}^{\pi} f(\theta) p_k(\theta) d\theta = 0 \quad \text{for all } k.$$

This proves the theorem for real-valued f .

Now suppose f is complex-valued. Write

$$f(\theta) = u(\theta) + iv(\theta),$$

where u and v are real-valued. Then

$$u(\theta) = \frac{f(\theta) + \overline{f(\theta)}}{2}, \quad v(\theta) = \frac{f(\theta) - \overline{f(\theta)}}{2i}.$$

Using

$$\widehat{\overline{f}}(n) = \overline{\widehat{f}(-n)},$$

we see that the Fourier coefficients of both u and v vanish. Since f is continuous at θ_0 , so are u and v . Applying the real-valued case to each of them gives

$$u(\theta_0) = v(\theta_0) = 0.$$

Hence

$$f(\theta_0) = 0.$$

□

2 Convolutions

Given two 2π -periodic integrable functions f and g on $\mathbb{R}$, we define their convolution on $[-\pi, \pi]$ by

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x - y) dy.$$

Since the functions are periodic, a change of variables also gives the equivalent formula

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) g(y) dy.$$

Loosely speaking, convolution behaves like a weighted average. For example, if $g \equiv 1$, then

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) dy,$$

so $f * g$ is constant and equal to the average value of f on the circle.

In the context of Fourier series, convolutions arise because the partial sums of the Fourier series of f can be written as

$$S_N(f)(x) = \sum_{n=-N}^N \widehat{f}(n) e^{inx}$$

and hence

$$\begin{aligned} S_N(f)(x) &= \sum_{n=-N}^N \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \right) e^{inx} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(\sum_{n=-N}^N e^{in(x-y)} \right) dy \\ &= (f * D_N)(x), \end{aligned}$$

where

$$D_N(x) = \sum_{n=-N}^N e^{inx}$$

is the N th Dirichlet kernel. Thus the problem of understanding the partial sums $S_N(f)$ reduces to understanding the convolution $f * D_N$.

Proposition 2.1. *Suppose that f , g , and h are 2π -periodic integrable functions. Then:*

- (i) $f * (g + h) = (f * g) + (f * h)$.
- (ii) $(cf) * g = c(f * g) = f * (cg)$ for any $c \in \mathbb{C}$.
- (iii) $f * g = g * f$.
- (iv) $(f * g) * h = f * (g * h)$.
- (v) $f * g$ is continuous.
- (vi)

$$\widehat{f * g}(n) = \widehat{f}(n) \widehat{g}(n) \quad \text{for all } n \in \mathbb{Z}.$$

Proof. Properties (i) and (ii) follow immediately from the linearity of the integral.

We first prove the remaining statements under the additional assumption that f , g , and h are continuous. Since they are also 2π -periodic, all of them are bounded on $\mathbb{R}$.

To prove (vi), we compute

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(x) e^{-inx} dx.$$

Substituting the definition of convolution gives

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x - y) dy \right) e^{-inx} dx.$$

Since the integrand is continuous, we may interchange the order of integration:

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} g(x - y) e^{-inx} dx \right) e^{-iny} dy.$$

Now make the change of variables $u = x - y$. By periodicity of $g(u)e^{-inu}$, we get

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} g(x - y) e^{-in(x-y)} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u) e^{-inu} du = \widehat{g}(n).$$

Therefore,

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} \widehat{g}(n) dy = \widehat{f}(n) \widehat{g}(n).$$

To prove (iii), we use the elementary fact that if F is continuous and 2π -periodic, then

$$\int_{-\pi}^{\pi} F(y) dy = \int_{-\pi}^{\pi} F(x - y) dy \quad \text{for every } x \in \mathbb{R}.$$

Indeed, this follows from a change of variables together with periodicity. Applying this with

$$F(y) = f(y)g(x - y),$$

we obtain

$$\begin{aligned} (f * g)(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x - y) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x - y) g(y) dy \\ &= (g * f)(x). \end{aligned}$$

To prove (iv), write

$$((f * g) * h)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(t) h(x - t) dt.$$

Substituting the definition of $f * g$ gives

$$((f * g) * h)(x) = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} f(y) g(t - y) h(x - t) dy dt.$$

Since the integrand is continuous, we may interchange the order of integration:

$$((f * g) * h)(x) = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} f(y) \left(\int_{-\pi}^{\pi} g(t - y) h(x - t) dt \right) dy.$$

Now let $u = t - y$. By periodicity,

$$\int_{-\pi}^{\pi} g(t - y)h(x - t) dt = \int_{-\pi}^{\pi} g(u)h((x - y) - u) du.$$

Hence

$$((f * g) * h)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)(g * h)(x - y) dy = (f * (g * h))(x).$$

Finally, we prove (v). Let $x_1, x_2 \in \mathbb{R}$. Then

$$(f * g)(x_1) - (f * g)(x_2) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)(g(x_1 - y) - g(x_2 - y)) dy.$$

Since g is continuous and 2π -periodic, it is uniformly continuous on $\mathbb{R}$. Thus for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|g(s) - g(t)| < \varepsilon \quad \text{whenever } |s - t| < \delta.$$

If $|x_1 - x_2| < \delta$, then for every y ,

$$|(x_1 - y) - (x_2 - y)| < \delta,$$

so

$$|g(x_1 - y) - g(x_2 - y)| < \varepsilon.$$

Therefore,

$$\begin{aligned} |(f * g)(x_1) - (f * g)(x_2)| &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(y)| |g(x_1 - y) - g(x_2 - y)| dy \\ &\leq \frac{\varepsilon}{2\pi} \int_{-\pi}^{\pi} |f(y)| dy. \end{aligned}$$

This shows that $f * g$ is continuous.

This completes the proof in the continuous case.

For general integrable functions, one uses the approximation lemma stated next: continuous 2π -periodic functions approximate integrable ones sufficiently well so that the identities above pass to the limit. Hence (iii)–(vi) remain valid for arbitrary 2π -periodic integrable functions. $\square$

3 Good Kernels

In the proof of Theorem 2.1, we constructed a sequence of trigonometric polynomials $\{p_k\}$ that become increasingly concentrated near the origin, allowing us to detect the behavior of a function near a single point. This suggests a more general principle: if a family of kernels concentrates its mass near 0, then convolution with those kernels should recover the local behavior of a function.

The family $\{p_k\}$ from Theorem 2.1 is a motivation for what follows. In general, it is not a family of good kernels, since the functions p_k are not normalized to have mean value 1. For the applications we have in mind, especially in Fourier analysis, this normalization is essential.

Recall from Section 3 that the N th partial sum of the Fourier series of f can be written as

$$S_N(f) = f * D_N,$$

where D_N is the N th Dirichlet kernel. Thus questions about convergence of Fourier series can be reformulated as questions about convolution with families of kernels. This motivates the following definition.

Definition 3.1. A family of kernels $\{K_n\}_{n=1}^{\infty}$ on the circle is called a good kernel if the following properties hold:

(a) For every $n \geq 1$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x) dx = 1.$$

(b) There exists a constant $M > 0$ such that for every $n \geq 1$,

$$\int_{-\pi}^{\pi} |K_n(x)| dx \leq M.$$

(c) For every $\delta > 0$,

$$\int_{\delta \leq |x| \leq \pi} |K_n(x)| dx \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Remark 3.2. The Dirichlet kernels $\{D_N\}$ do not form a family of good kernels. Although

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(x) dx = 1,$$

they fail the uniform L^1 bound required in property (b). In fact,

$$\int_{-\pi}^{\pi} |D_N(x)| dx$$

grows like a constant multiple of $\log N$. Thus convolution with D_N is not controlled in the same way as convolution with a good kernel.

This is one reason why pointwise convergence of the ordinary partial sums

$$S_N(f) = f * D_N$$

is more delicate than convergence of the averaged sums. It motivates replacing ordinary convergence by summability methods, such as Cesàro summability with the Fejér kernel and Abel summability with the Poisson kernel. It also points toward other notions of convergence, such as L^2 convergence, where Fourier partial sums behave much better.

Property (a) says that each kernel has total mass 1, so convolution with K_n acts like an averaging process. Property (b) gives a uniform L^1 bound, which keeps these averages under control. Property (c) says that, away from the origin, the mass of K_n becomes negligible as n grows. Taken together, these conditions express the idea that the kernels become concentrated near 0 while preserving total mass.

Lemma 3.3. Let $f \in L^1([-\pi, \pi])$, and let x_0 be a point at which f is continuous. Then for every $\varepsilon > 0$, there exists a continuous 2π -periodic function g such that

$$\int_{-\pi}^{\pi} |f(y) - g(y)| dy < \varepsilon$$

and

$$g(x_0) = f(x_0).$$

Remark 3.4. The lemma is a consequence of the density of continuous functions in $L^1([-\pi, \pi])$, together with a local adjustment near the point x_0 .

In practice, we often encounter families for which $K_n(x) \geq 0$. In that case, property (b) follows immediately from (a). Intuitively, the kernels K_n behave like weight distributions on the circle: property (a) says that each K_n has total mass 1, while property (c) says that this mass concentrates near the origin as n becomes large.

The importance of good kernels comes from their interaction with convolutions.

Theorem 3.5. *Let $\{K_n\}_{n=1}^\infty$ be a family of good kernels, and let f be an integrable function on the circle. Then*

$$\lim_{n \rightarrow \infty} (f * K_n)(x) = f(x)$$

whenever f is continuous at x .

If f is continuous everywhere on the circle, then the convergence is uniform.

Proof. Fix a point x at which f is continuous, and let $\varepsilon > 0$ be given. By continuity of f at x , there exists $\delta > 0$ such that

$$|y| < \delta \implies |f(x - y) - f(x)| < \varepsilon.$$

Using property (a) of good kernels, we may write

$$\begin{aligned} (f * K_n)(x) - f(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x - y) K_n(y) dy - f(x) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x - y) K_n(y) dy - \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) K_n(y) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) (f(x - y) - f(x)) dy. \end{aligned}$$

Therefore,

$$\begin{aligned} |(f * K_n)(x) - f(x)| &\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |K_n(y)| |f(x - y) - f(x)| dy \\ &= \frac{1}{2\pi} \int_{|y| < \delta} |K_n(y)| |f(x - y) - f(x)| dy \\ &\quad + \frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| |f(x - y) - f(x)| dy. \end{aligned}$$

On the region $|y| < \delta$, we have

$$|f(x - y) - f(x)| < \varepsilon,$$

so

$$\begin{aligned} \frac{1}{2\pi} \int_{|y| < \delta} |K_n(y)| |f(x - y) - f(x)| dy &\leq \frac{\varepsilon}{2\pi} \int_{-\pi}^{\pi} |K_n(y)| dy \\ &\leq \frac{M\varepsilon}{2\pi}, \end{aligned}$$

by property (b).

Since f is integrable on the circle, it is bounded on the compact interval $[-\pi, \pi]$. Choose $B > 0$ such that

$$|f(t)| \leq B \quad \text{for all } t \in [-\pi, \pi].$$

Then

$$|f(x - y) - f(x)| \leq 2B,$$

and hence

$$\frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| |f(x - y) - f(x)| dy \leq \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy.$$

By property (c), this term tends to 0 as $n \rightarrow \infty$.

Combining the two estimates, we obtain

$$|(f * K_n)(x) - f(x)| \leq \frac{M\varepsilon}{2\pi} + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy.$$

Taking lim sup as $n \rightarrow \infty$ gives

$$\limsup_{n \rightarrow \infty} |(f * K_n)(x) - f(x)| \leq \frac{M\varepsilon}{2\pi}.$$

Since $\varepsilon > 0$ was arbitrary, it follows that

$$(f * K_n)(x) \rightarrow f(x) \quad \text{as } n \rightarrow \infty.$$

Now assume that f is continuous everywhere on the circle. Since the circle is compact, f is uniformly continuous. Therefore, for the given $\varepsilon > 0$, we may choose $\delta > 0$ so that

$$|y| < \delta \implies |f(x - y) - f(x)| < \varepsilon \quad \text{for every } x.$$

Repeating the argument above, we obtain

$$|(f * K_n)(x) - f(x)| \leq \frac{M\varepsilon}{2\pi} + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy$$

for every x . Taking the supremum over x gives

$$\sup_x |(f * K_n)(x) - f(x)| \leq \frac{M\varepsilon}{2\pi} + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy.$$

Now let $n \rightarrow \infty$. By property (c),

$$\limsup_{n \rightarrow \infty} \sup_x |(f * K_n)(x) - f(x)| \leq \frac{M\varepsilon}{2\pi}.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that

$$\sup_x |(f * K_n)(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence the convergence is uniform. The Dirichlet kernels are not a family of good kernels, not because uniform convergence would fail by itself, but because they do not satisfy the uniform L^1 bound required in property (b).

□

4 Cesàro and Abel Summability: Applications to Fourier Series

Since Fourier series may fail to converge at individual points, we can try to recover the function by interpreting ordinary convergence in a weaker sense.

4.1 Cesàro Means and Summation

Suppose we are given a series of complex numbers

$$\sum_{k=0}^{\infty} c_k.$$

Its n th partial sum is

$$s_n = \sum_{k=0}^n c_k.$$

Ordinary convergence means that $s_n \rightarrow s$ as $n \rightarrow \infty$.

A basic example is the alternating series

$$1 - 1 + 1 - 1 + \cdots = \sum_{k=0}^{\infty} (-1)^k, \quad (1)$$

whose partial sums are

$$1, 0, 1, 0, \dots$$

and therefore (1) does not converge. Nevertheless, these partial sums oscillate evenly between 1 and 0, so it is natural to regard their average as the more meaningful limiting value.

Definition 4.1. Given a sequence of partial sums $\{s_k\}$, the N -th *Cesàro mean* is defined by

$$\sigma_N = \frac{s_0 + s_1 + \cdots + s_{N-1}}{N}.$$

If $\sigma_N \rightarrow \sigma$ as $N \rightarrow \infty$, we say that the series

$$\sum_{k=0}^{\infty} c_k$$

is *Cesàro summable* to σ .

For the alternating series above, one checks directly that

$$\sigma_N \rightarrow \frac{1}{2}.$$

Thus that series is Cesàro summable to $1/2$.

Theorem 4.2. *If the series*

$$\sum_{k=0}^{\infty} c_k$$

converges in the usual sense to s , then it is also Cesàro summable to s .

Proof. Let

$$s_n = \sum_{k=0}^n c_k$$

denote the n th partial sum, and suppose that

$$s_n \rightarrow s \quad \text{as } n \rightarrow \infty.$$

By definition, the N th Cesàro mean is

$$\sigma_N = \frac{s_0 + s_1 + \cdots + s_{N-1}}{N}.$$

We must show that $\sigma_N \rightarrow s$.

Fix $\varepsilon > 0$. Since $s_n \rightarrow s$, there exists an integer N_0 such that

$$|s_n - s| < \varepsilon \quad \text{whenever } n \geq N_0.$$

Then for $N > N_0$, we write

$$\begin{aligned} |\sigma_N - s| &= \left| \frac{1}{N} \sum_{k=0}^{N-1} s_k - s \right| \\ &= \left| \frac{1}{N} \sum_{k=0}^{N-1} (s_k - s) \right| \\ &\leq \frac{1}{N} \sum_{k=0}^{N-1} |s_k - s| \\ &= \frac{1}{N} \sum_{k=0}^{N_0-1} |s_k - s| + \frac{1}{N} \sum_{k=N_0}^{N-1} |s_k - s|. \end{aligned}$$

For the second sum, each term is at most ε , so

$$\frac{1}{N} \sum_{k=N_0}^{N-1} |s_k - s| \leq \frac{N - N_0}{N} \varepsilon \leq \varepsilon.$$

For the first sum, the number of terms is fixed, so if we let

$$C = \sum_{k=0}^{N_0-1} |s_k - s|,$$

then

$$\frac{1}{N} \sum_{k=0}^{N_0-1} |s_k - s| = \frac{C}{N} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Hence, for all sufficiently large N ,

$$|\sigma_N - s| \leq \frac{C}{N} + \varepsilon < 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that

$$\sigma_N \rightarrow s.$$

Therefore the series is Cesàro summable to s . □

4.2 Fejérs Theorem

We now apply the idea of Cesàro summation to Fourier series.

Recall that

$$S_n(f) = f * D_n,$$

where D_n is the n th Dirichlet kernel. Define the N th Cesàro mean of the Fourier series of f by

$$\sigma_N(f)(x) = \frac{S_0(f)(x) + S_1(f)(x) + \cdots + S_{N-1}(f)(x)}{N}.$$

Since convolution is linear,

$$\sigma_N(f)(x) = (f * F_N)(x),$$

where

$$F_N(x) = \frac{D_0(x) + D_1(x) + \cdots + D_{N-1}(x)}{N}$$

is called the N th *Fejér kernel*.

Lemma 4.3. *For every $N \geq 1$,*

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)} \quad (x \neq 0),$$

and this expression extends continuously to $x = 0$ by setting

$$F_N(0) = N.$$

Moreover, the family $\{F_N\}$ is a family of good kernels.

Proof. For $x \neq 0$, the identity

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$$

is a standard trigonometric computation obtained from the formula for the Dirichlet kernels. At $x = 0$, one has

$$F_N(0) = \frac{D_0(0) + \cdots + D_{N-1}(0)}{N} = \frac{1 + 3 + \cdots + (2N-1)}{N} = N,$$

so the formula has a removable singularity at $x = 0$. Thus F_N is well defined on the whole circle.

We now verify that $\{F_N\}$ is a family of good kernels.

First, since each D_n satisfies

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} D_n(x) dx = 1,$$

we also have

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N(x) dx = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{2\pi} \int_{-\pi}^{\pi} D_n(x) dx = 1.$$

Next, for $x \neq 0$ the explicit formula shows that

$$F_N(x) \geq 0,$$

since both numerator and denominator are squares, and we have already seen that $F_N(0) = N > 0$. Therefore

$$\int_{-\pi}^{\pi} |F_N(x)| dx = \int_{-\pi}^{\pi} F_N(x) dx = 2\pi,$$

so the family is uniformly bounded in L^1 .

Finally, fix $\delta > 0$. On the closed set

$$E_\delta = \{x \in [-\pi, \pi] : \delta \leq |x| \leq \pi\},$$

the function $x \mapsto \sin^2(x/2)$ is continuous and strictly positive, since its only zeros occur at integer multiples of 2π , and $0 \notin E_\delta$. Because E_δ is compact, $\sin^2(x/2)$ attains a positive minimum there. Thus there exists a constant $c_\delta > 0$ such that

$$\sin^2(x/2) \geq c_\delta > 0 \quad \text{for all } x \text{ with } \delta \leq |x| \leq \pi.$$

Hence, whenever $\delta \leq |x| \leq \pi$,

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)} \leq \frac{1}{N} \frac{1}{c_\delta} = \frac{1}{Nc_\delta},$$

since $\sin^2(Nx/2) \leq 1$. Therefore,

$$\int_{\delta \leq |x| \leq \pi} |F_N(x)| dx \leq \int_{\delta \leq |x| \leq \pi} \frac{1}{Nc_\delta} dx \leq \frac{2\pi}{Nc_\delta} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Since this holds for every $\delta > 0$, property (c) is verified. Therefore $\{F_N\}$ is a family of good kernels. □

Theorem 4.4 (Fejér's Theorem). *If f is integrable on the circle, then the Fourier series of f is Cesàro summable to f at every point of continuity of f .*

Moreover, if f is continuous on the circle, then the Fourier series of f is uniformly Cesàro summable to f .

Proof. Since

$$\sigma_N(f) = f * F_N$$

and the Fejér kernels form a family of good kernels, the conclusion follows immediately from Theorem 4.1. □

Corollary 4.5. *If f is integrable on the circle and*

$$\widehat{f}(n) = 0 \quad \text{for all } n \in \mathbb{Z},$$

then $f = 0$ at every point of continuity of f .

Proof. If all Fourier coefficients vanish, then every partial sum $S_N(f)$ is identically zero. Hence every Cesàro mean $\sigma_N(f)$ is also identically zero. By Fejér's theorem, $\sigma_N(f)(x) \rightarrow f(x)$ at every point of continuity of f , so $f(x) = 0$ at each such point. □

Corollary 4.6. *Every continuous function on the circle can be uniformly approximated by trigonometric polynomials.*

Proof. If f is continuous, then by Fejér's theorem,

$$\sigma_N(f) \rightarrow f$$

uniformly on the circle. But each $\sigma_N(f)$ is a finite average of partial sums of the Fourier series, and each partial sum is a trigonometric polynomial. Therefore each $\sigma_N(f)$ is itself a trigonometric polynomial. Hence f is the uniform limit of trigonometric polynomials. $\square$

4.3 Abel Means and Summation

There is another method of summation, due to Abel, which is even more powerful than Cesàro summation.

Definition 4.7. A series

$$\sum_{k=0}^{\infty} c_k$$

is said to be *Abel summable* to s if for every $0 \leq r < 1$, the power series

$$A(r) = \sum_{k=0}^{\infty} c_k r^k$$

converges, and

$$\lim_{r \rightarrow 1^-} A(r) = s.$$

The quantities $A(r)$ are called the *Abel means* of the series.

As in the Cesàro case, ordinary convergence implies Abel summability to the same limit. In fact Abel summability is strictly stronger than Cesàro summability.

For example,

$$1 - 2 + 3 - 4 + 5 - \cdots = \sum_{k=0}^{\infty} (-1)^k (k+1)$$

is Abel summable to $1/4$, since

$$\sum_{k=0}^{\infty} (-1)^k (k+1) r^k = \frac{1}{(1+r)^2},$$

and therefore

$$\lim_{r \rightarrow 1^-} \sum_{k=0}^{\infty} (-1)^k (k+1) r^k = \frac{1}{4}.$$

4.4 The Poisson Kernel and Dirichlets Problem in the Unit Disc

To adapt Abel summability to Fourier series, define the Abel means of the Fourier series

$$f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$$

by

$$A_r(f)(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} a_n e^{in\theta}, \quad 0 \leq r < 1.$$

Since f is integrable, the Fourier coefficients a_n are bounded, so this series converges absolutely and uniformly for each fixed $r < 1$.

The key point is that these Abel means may also be written as convolutions. Define the *Poisson kernel* by

$$P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta}.$$

Then

$$A_r(f)(\theta) = (f * P_r)(\theta).$$

Indeed,

$$\begin{aligned} A_r(f)(\theta) &= \sum_{n=-\infty}^{\infty} r^{|n|} a_n e^{in\theta} \\ &= \sum_{n=-\infty}^{\infty} r^{|n|} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) e^{-in\varphi} d\varphi \right) e^{in\theta} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) \left(\sum_{n=-\infty}^{\infty} r^{|n|} e^{-in(\varphi-\theta)} \right) d\varphi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) P_r(\theta - \varphi) d\varphi \\ &= (f * P_r)(\theta), \end{aligned}$$

where the interchange of sum and integral is justified by uniform convergence.

Lemma 4.8. *If $0 \leq r < 1$, then*

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}.$$

Moreover, as $r \rightarrow 1^-$, the Poisson kernels form a family of good kernels.

Proof. The identity

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}$$

is the standard closed form for the Poisson kernel.

We now check that $\{P_r\}_{0 \leq r < 1}$ is a family of good kernels as $r \rightarrow 1^-$.

First,

$$P_r(\theta) \geq 0$$

for all θ , since the numerator is positive and the denominator is positive.

Also, integrating the defining Fourier series term by term gives

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta) d\theta = 1,$$

because all nonconstant exponential terms integrate to zero.

Since $P_r \geq 0$, it follows immediately that

$$\int_{-\pi}^{\pi} |P_r(\theta)| d\theta = \int_{-\pi}^{\pi} P_r(\theta) d\theta = 2\pi,$$

so property (b) holds.

Finally, fix $\delta > 0$. We use the identity

$$1 - 2r \cos \theta + r^2 = (1 - r)^2 + 2r(1 - \cos \theta).$$

Let

$$E_\delta = \{\theta \in [-\pi, \pi] : \delta \leq |\theta| \leq \pi\}.$$

The function $\theta \mapsto 1 - \cos \theta$ is continuous and vanishes only at $\theta = 0$. Since $0 \notin E_\delta$, it follows that $1 - \cos \theta$ is strictly positive on E_δ . Because E_δ is compact, the function $1 - \cos \theta$ attains a positive minimum there. Therefore there exists a constant $c_\delta > 0$ such that

$$1 - \cos \theta \geq c_\delta \quad \text{whenever } \delta \leq |\theta| \leq \pi.$$

Hence, if $1/2 \leq r < 1$ and $\delta \leq |\theta| \leq \pi$, then

$$1 - 2r \cos \theta + r^2 = (1 - r)^2 + 2r(1 - \cos \theta) \geq 2r c_\delta \geq c_\delta.$$

Therefore

$$P_r(\theta) \leq \frac{1 - r^2}{c_\delta} \quad \text{whenever } \delta \leq |\theta| \leq \pi.$$

It follows that

$$\int_{\delta \leq |\theta| \leq \pi} |P_r(\theta)| d\theta \leq \frac{1 - r^2}{c_\delta} \cdot 2\pi \rightarrow 0 \quad \text{as } r \rightarrow 1^-.$$

Since this holds for every $\delta > 0$, property (c) is verified.

Thus the Poisson kernels form a family of good kernels. □

Theorem 4.9. *The Fourier series of an integrable function on the circle is Abel summable to f at every point of continuity of f .*

Moreover, if f is continuous on the circle, then the Fourier series is uniformly Abel summable to f .

Proof. Since

$$A_r(f) = f * P_r$$

and the Poisson kernels form a family of good kernels as $r \rightarrow 1^-$, the result follows directly from Theorem 4.1. □

We now return to the Dirichlet problem in the unit disc. Motivated by the Fourier expansion, define

$$u(r, \theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta},$$

where a_m is the m th Fourier coefficient of f . Equivalently,

$$u(r, \theta) = A_r(f)(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\varphi) P_r(\theta - \varphi) d\varphi.$$

Theorem 4.10. *Let f be an integrable function on the unit circle, and define*

$$u(r, \theta) = (f * P_r)(\theta).$$

Then:

(i) *u has two continuous derivatives in the unit disc and satisfies*

$$\Delta u = 0.$$

(ii) *If θ is a point of continuity of f , then*

$$\lim_{r \rightarrow 1^-} u(r, \theta) = f(\theta).$$

If f is continuous everywhere, then the convergence is uniform in θ .

(iii) *If f is continuous, then u is the unique solution of the Dirichlet problem in the disc with boundary values f .*

Proof. For part (i), recall that

$$u(r, \theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}.$$

Fix $\rho < 1$. On the closed disc $0 \leq r \leq \rho$, the series for u and all of its termwise derivatives converge absolutely and uniformly, because the factors $r^{|m|}$ give geometric decay. Therefore u may be differentiated term by term, in fact infinitely many times, inside the unit disc.

In polar coordinates,

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$

Applying this operator term by term to

$$r^{|m|} e^{im\theta}$$

shows that each term is harmonic, hence

$$\Delta u = 0.$$

□

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